

What Goes Wrong: A Theory of Departures

Parts II to IV derived everything from a short list of assumptions: the mean is $\mathbf{X}\boldsymbol{\beta}$ for the regressors actually used, the errors are uncorrelated with a common variance, they are normal, and the regressors are fixed and measured without error. Real data meet each of these only approximately. This chapter asks, one assumption at a time, what a specific departure does to the least squares estimator, to s^2 and to the tests and intervals built from them. Omitting a regressor biases the coefficients, and yet the biased estimator can still have the smaller mean squared error. A wrong covariance matrix leaves the estimator nearly efficient but can make its standard error wrong by a factor of two or more. With nonnormal errors the t and F tests are asymptotically valid when no observation has large leverage, while tests and intervals for σ^2 are not valid in any sample size. A single bad observation can move the estimate arbitrarily far. With random regressors and a curved mean, the usual standard errors describe the wrong target. Collinearity breaks no assumption at all, but it inflates variances. Chapters 20–22 turn to diagnosis and remedies, and Chapter 23 to inference that relies less on the assumptions.

What you need. Chapter 5 (the moments of $\hat{\boldsymbol{\beta}}$ under departures, Proposition 5.5.7, and omitted regressors, Proposition 5.7.2), Chapter 6 (the Frisch–Waugh–Lovell theorem, leverage, Kruskal’s theorem and the population projection), Chapter 7 (Gauss–Markov and Aitken), Chapter 10 (the deletion formulas, Proposition 10.6.2), and the tests and intervals of Chapter 11 and Chapter 12. Section 14.1 treats random regressors under the normal model, and Section 19.5 builds on it. Two earlier chapters carry more of the weight than the list above suggests. Chapter 2 supplies the moments recomputed under every departure — the covariance of a linear function, the mean and variance of a quadratic form, the expected residual sum of squares and the law of total covariance (Theorem 2.2.1, Theorem 2.3.1, Theorem 2.3.3, Eq. 2.3.1, Proposition 2.5.4) — and Chapter 1 the spectral theorem and extremal Rayleigh quotient that Sections 19.2 and 19.6 measure efficiency with, together with the QR factorization and the Cauchy–Schwarz inequality (Theorem 1.7.2, Theorem 1.7.7, Theorem 1.5.2, Proposition 1.1.1). Section 19.2 points forward to feasible generalized least squares (Section 31.4); that is a promise, not a prerequisite.

19.-1.1 Roadmap

- [Omitting and adding regressors](#)
- [An incorrect covariance matrix](#)
- [Nonnormal errors](#)
- [A single bad observation](#)
- [Random regressors and a curved mean](#)
- [Collinearity, previewed](#)
- [Summary and notes](#)